

InvoiceFlow

1. What I Built

InvoiceFlow is a payment reminder web app for small businesses. It lets an admin track customers, manage invoices across their full lifecycle, and send payment reminder emails — all from one place. The app is fully deployed and works on desktop and mobile.

The project covers all six required areas from the assignment: invoices, reminders, real email, search and filtering, a dashboard, and a responsive UI. A few extras were added that felt natural to the product, explained in Section 3.

2. Features

Dashboard

The first thing the admin sees when they open the app. It shows total revenue collected, total outstanding amount, total overdue, and the five most recent invoices. The goal was that the admin should understand the state of their business the moment they open the app, without clicking anything.

Customers

A directory of all clients the admin invoices. Each customer has a profile showing their contact details, every invoice linked to them, and a total of how much they have paid versus how much is still owed. When creating a new customer, the admin can create an invoice at the same time to reduce extra steps.

Invoices

The core of the app. Admins can create, view, edit, and delete invoices. Each invoice has line items, a due date, a tax rate, and a status. Totals are always calculated on the server — the admin submits line items and the server works out the subtotal, tax, and total. Invoice numbers are auto-generated (INV-0001, INV-0002, etc.).

Each invoice moves through a clear lifecycle:

Status	What it means
Draft	Created but not yet sent to the customer
Sent	Emailed to the customer
Pending	Waiting for payment

Paid	Payment received
Overdue	Past due date, still unpaid
Cancelled	No longer active

Invoices are automatically marked overdue when their due date passes — no manual update needed. The list can be filtered by status and searched by customer name or invoice number.

Reminders

A queue of invoices that need a follow-up. The admin can send a reminder to one customer at a time — with the option to edit the subject and message before sending — or bulk send to all eligible invoices at once. Every reminder records when it was sent, and if the customer opens the email, that is recorded too.

Real Email

Emails are sent through Resend, a transactional email service. The email includes the invoice details and amount owed. As the free tier of Resend only allows sending to the verified owner's email address, all reminder emails are currently delivered to my own inbox rather than the actual customers. API keys are stored in environment variables and never committed to the repository.

Search and Filtering

Invoices can be searched by customer name or invoice number, and filtered by status. Both work together so the admin can find exactly what they need quickly. Customers can also be searched by name. The idea was that no admin should ever have to scroll through a long list.

Responsive UI

The app works on desktop and mobile. The sidebar collapses into a hamburger menu on small screens. The admin can switch between light and dark mode — the choice is saved and remembered across sessions.

3. What I Added Beyond the Brief

Connection of Reminder and Invoice

Every reminder is linked directly to an invoice. When you send a reminder, it pulls the invoice details (amount, due date, customer) automatically. The reminder knows which invoice it belongs to, so the admin never has to manually match them up.

Separate Customer Section

Customers are their own section, not just a field inside an invoice. This means you can see a full profile per customer — all their invoices together, total paid, total still owed — rather than hunting through individual invoices to piece that together.

Sending All Pending Invoices at Once

On the reminders page there is a bulk send button. Instead of going one by one, the admin can send reminders to every eligible invoice in a single click. Before sending, the subject and message can still be edited. This saves time when many customers owe at the same time.

Keeping Track of Last Sent Reminder

Every time a reminder is sent, the app logs the exact time it was sent. If the customer opens the email, that time is also recorded via a webhook from Resend. So the admin can see not just when they sent it, but whether the customer actually opened it — useful for deciding whether to follow up again.

4. Tech Stack

Area	Technology
Frontend	React 19 + Vite, React Router
Styling	Bootstrap 5 + CSS custom properties
Backend	Node.js + Express
Database	MongoDB via Mongoose
Email	Resend API
Frontend hosting	Vercel
Backend hosting	Railway

5. Decisions I Made

Invoice Totals Are Calculated on the Server

The admin submits line items and the server calculates the total. This keeps numbers correct and consistent.

All API Calls Go Through One Place

Every network request goes through a single `api/` folder. If the backend changes, there is only one place to update.

Overdue Invoices Update Automatically

When the reminders page loads, invoices past their due date are automatically marked overdue. No manual update needed.

No Passwords or Keys in the Code

Sensitive values like the database password and API key are stored in environment variables, never in the code. A `.env.example` file is included so anyone setting up the project knows what they need.

6. What I Would Add Next

- PDF export of the dashboard so the admin can save or share a financial summary
- A settings page for configuring business name, default tax rate, and email templates